

PDF Report Enhancement V3 - Health Score Explainability

Objective

Improve the Electrical Health Score section of the PDF report by making the score transparent, understandable, and actionable.

Currently the report displays a Health Score and category breakdown.

Example:

Health Score: 70/100

Load Utilization: 10/30

Energy Efficiency: 20/30

Safety Readiness: 20/20

Solar Readiness: 20/20

While useful, users still do not understand:

- Why points were deducted
- What contributed to the score
- How to improve the score

The goal is to transform the Health Score from a number into an explainable assessment.

Problem

Current experience:

Health Score = 70/100

User reaction:

Why not 100?

What reduced my score?

How can I improve it?

The PDF should answer these questions automatically.

New Section Structure

Electrical Health Assessment



Overall Score



Score Breakdown



Score Explanation



Improvement Opportunities

Section 1 - Overall Health Score

Keep existing score card.

Example:

Electrical Health Score

70 / 100

Status

GOOD

Status Rules

90-100

Excellent

75-89

Good

60-74

Fair

Below 60

Needs Attention

Display status badge near the score.

Section 2 - Score Breakdown

Keep existing category breakdown.

Example:

Load Utilization

10 / 30

Energy Efficiency

20 / 30

Safety Readiness

20 / 20

Solar Readiness

20 / 20

Add progress bars for visual clarity.

Use existing PDF design language.

Section 3 - Why This Score?

Add a new subsection.

Title:

Why This Score?

This section should automatically explain deductions.

Examples:

Load Utilization

High connected load concentration reduced the score.

Multiple appliances contribute significantly to total load.

Energy Efficiency

Several appliances consume above-average energy.

Optimization opportunities were identified.

Safety Readiness

Electrical recommendations are within acceptable residential standards.

No major safety concerns detected.

Solar Readiness

Current energy profile is suitable for solar integration.

Solar recommendation generated successfully.

Only show explanations relevant to the actual assessment.

Do not use generic filler text.

Section 4 - Improvement Opportunities

Add a new subsection.

Title:

How to Improve Your Score

Generate recommendations dynamically.

Examples:

Increase Score By Improving Energy Efficiency

- ✓ Replace older refrigerator models
- ✓ Reduce unnecessary standby consumption
- ✓ Upgrade inefficient appliances

Increase Score By Improving Load Balance

- ✓ Distribute appliance usage more evenly
- ✓ Avoid simultaneous operation of high-load devices

Increase Score By Improving Solar Readiness

- ✓ Consider recommended solar installation
- ✓ Monitor monthly energy usage trends

Each recommendation should be tied to the affected category.

Section 5 - Potential Score Improvement

Add a summary box.

Example:

Potential Health Score Improvement

Current Score

70 / 100

Potential Score

82 / 100

If recommended optimizations are implemented.

This is an estimate only.

Do not present as a guaranteed result.

UX Requirements

The Health Score section should answer:

What is my score?

Why did I get this score?

What reduced my score?

What can I improve?

How much improvement is possible?

Without requiring the user to interpret the report themselves.

Design Requirements

Maintain existing PDF styling.

Do not redesign the report.

Enhance the current Health Score section only.

Keep explanations concise.

Avoid large paragraphs.

Use:

- Bullet points
- Status indicators
- Progress bars
- Small callout boxes

for readability.

Success Criteria

After reading the Health Score section, users should fully understand:

- ✓ Why their score is not 100
- ✓ What factors contributed positively
- ✓ What factors reduced the score
- ✓ What actions can improve it
- ✓ The approximate improvement potential

The Health Score should feel like an intelligent consultant assessment rather than a simple numerical rating.